A learning system can connect observations to a useful internal representation. We study the visible structure of a document before interpreting the words on the page. This preserves illustrations with many small labels and keeps those labels attached to their original diagram. The boundary is a separate result.

The next paragraph begins with an indentation rather than a large blank line. Its final line is intentionally short. A reliable layout should not combine neighboring columns or extend a paragraph through the central gutter.

Text and drawings provide different kinds of evidence. An exported page may contain a reusable vector group, while a scanned page contains only one raster image. The visible organization is the same in both examples. Selection should follow content boundaries and should not depend on an incidental label.

The complete figure below contains four labeled panels. The panel letters, axes and labels are internal content. Its caption is ordinary prose placed below the artwork. The running title and page number are page furniture.

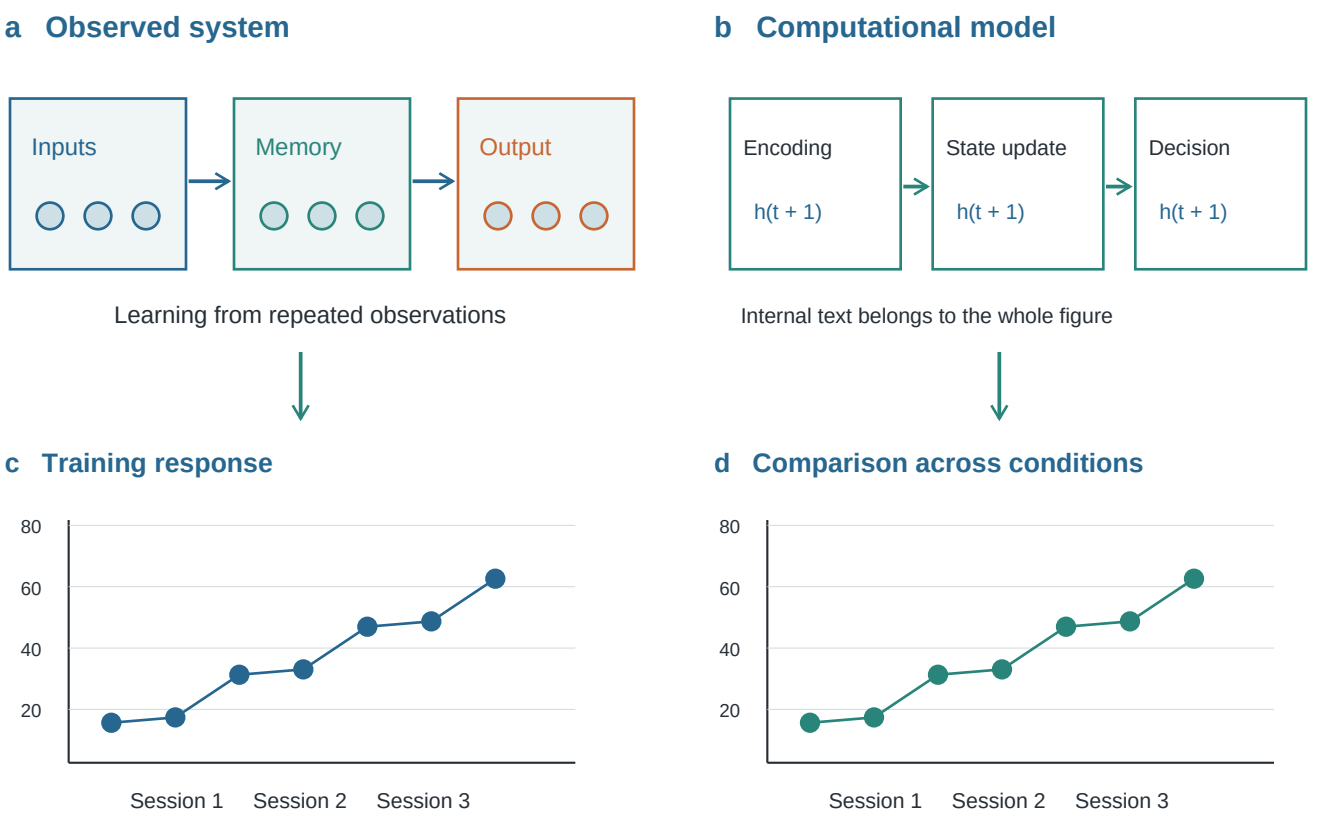

Fig. 1 | An independent diagram for layout validation. The upper panels show a chain of observed states and a computational counterpart. Small labels remain part of the complete illustration. No scientific result is claimed by this synthetic example.

The lower panels contain a deliberately simple series of points. Axes and session labels must not become individual selectable paragraphs. This column is the continuation of the caption and is outside the figure. All content on this page was written for testing.

The body resumes after the figure and its caption. This short paragraph provides a distinct lower block. A footer rule separates the page furniture below.

Results

A learning system can connect observations to a useful internal representation. We study the visible structure of a document before interpreting the words on the page. This preserves illustrations with many small labels and keeps those labels attached to their original diagram. The boundary is a separate result. The next paragraph begins with an indentation rather than a large blank line. Its final line is intentionally short.

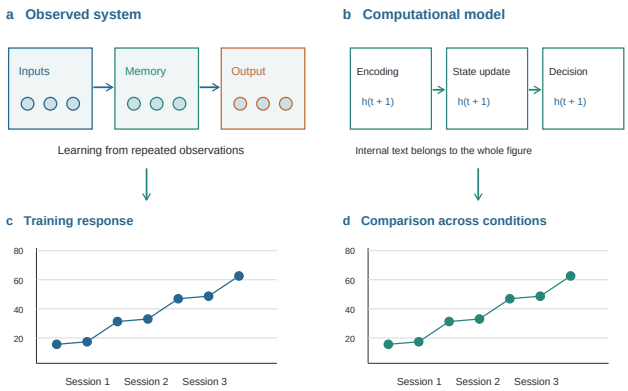

Fig. 2 | A smaller diagram in the left column. Adjacent body text belongs to a separate region.

The next paragraph begins with an indentation rather than a large blank line. Its final line is intentionally short. A reliable layout should not combine neighboring columns or extend a paragraph through the central gutter.

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Text and drawings provide different kinds of evidence. An exported page may contain a reusable vector group, while a scanned page contains only one raster image. The visible organization is the same in both examples. Selection should follow content boundaries and should not depend on an incidental label. The next paragraph begins with an indentation rather than a large blank line. Its final line is intentionally short. A reliable layout should not combine neighboring columns or extend a paragraph through the central gutter. This paragraph stands beside a compact illustration. Its column must remain separate from the diagram.

The next paragraph begins with an indentation rather than a large blank line. Its final line is intentionally short. A reliable layout should not combine neighboring columns or extend a paragraph through the central gutter.

$$s(t + 1) = a\, s(t) + b\, u(t) \tag{1}$$

Here s is a state and u is a measured input. This equation is included only to test spatial grouping. A formula is not a figure simply because it is short.

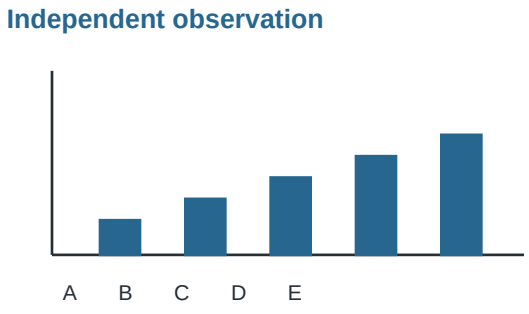

Fig. 3 | A separate synthetic bar chart. This caption defines a distinct local block.

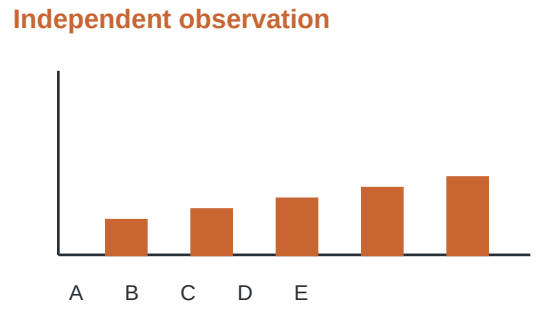

Fig. 4 | A separate synthetic bar chart. This caption defines a distinct local block.